That the inverse model is genuinely out of the loop is a verified property rather than a claim about intent. Under the direct-valve configuration the adaptation-site instrumentation reports zero valve elements attributable to the inverse model and zero adapter applications at the inverse-model site across a complete run. The actuator key recorded inside the checkpoints still reads `id_aam_fd`; it is retained so that previously trained checkpoints remain loadable and is a legacy identifier, not a description of the runtime path. The reported counters are `valve_elements` $= 0$, `adapter_applied_env_ticks` $= 0$ and `online_aam_residual_scale` $= 0.0$; were the inverse model in the path, all three would be non-zero.

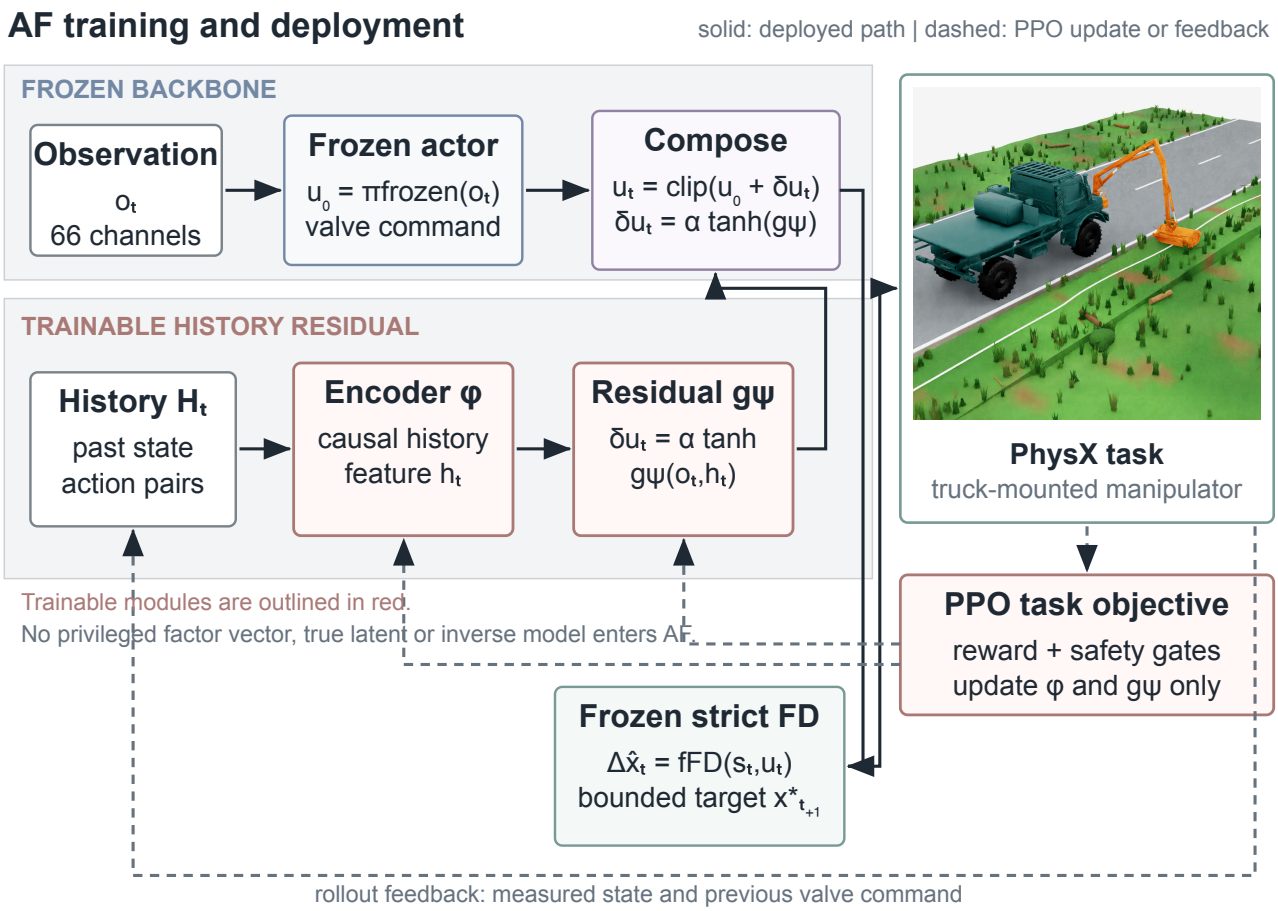


Figure 6.6: Training and deployment path of AF. The frozen base actor emits the valve command $\boldsymbol{u}_0$. A temporal encoder and residual head use causal state–action history to add the bounded correction of Equation (6.11). Only these two modules receive the PPO task gradient; the base actor and strict forward model remain frozen. The composed valve command drives the forward model and the shared PhysX verge-following task. Dashed paths denote training gradients and rollout feedback. No privileged factor vector, true latent or inverse model enters the AF command path.

6.7.1 Why the Inverse Model Is Omitted

The three stacks are not three attempts at one thing, and the inverse model's place in two of them follows from what each is for. The command ROM together with PhysX *is* the deployment plant (Section 5.4.1), and it is the only actuator representation calibrated